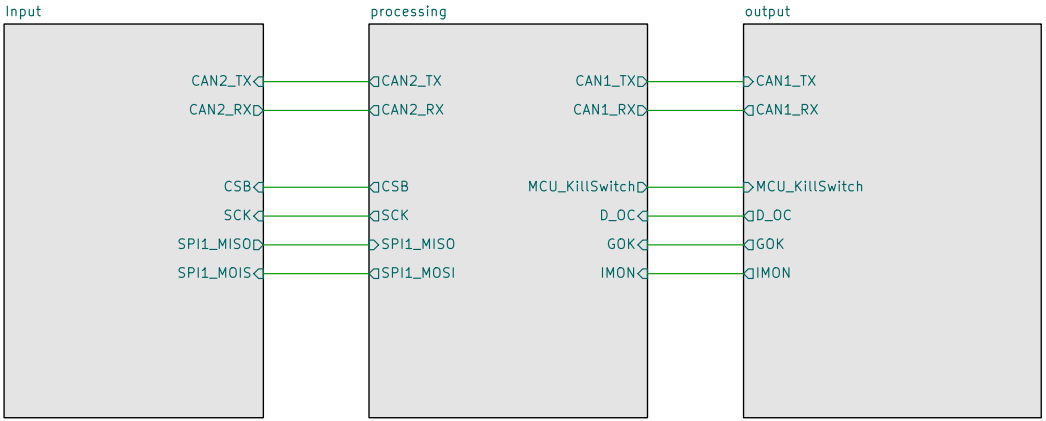


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Project: torpedokontroller

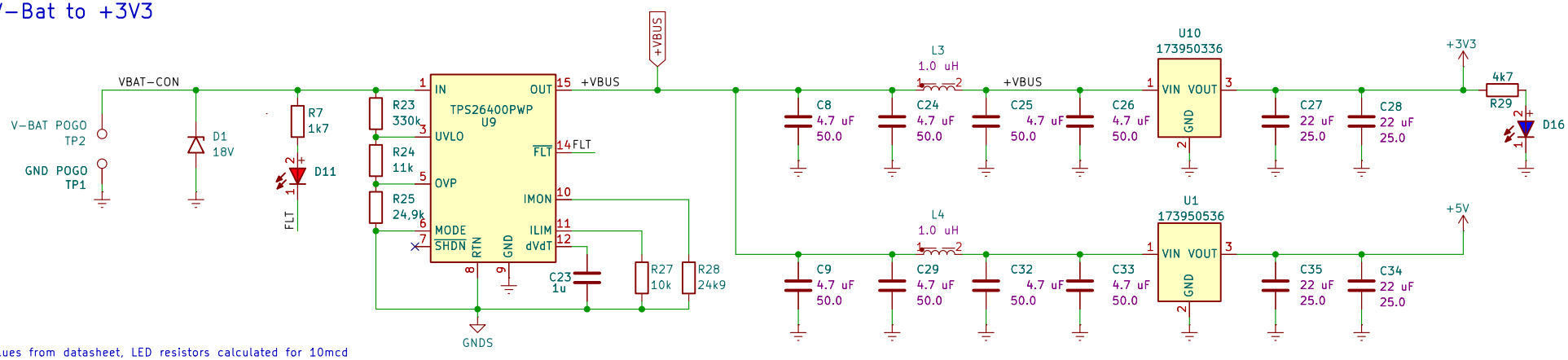
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File: torpedokontroller.kicad_sch

Title:

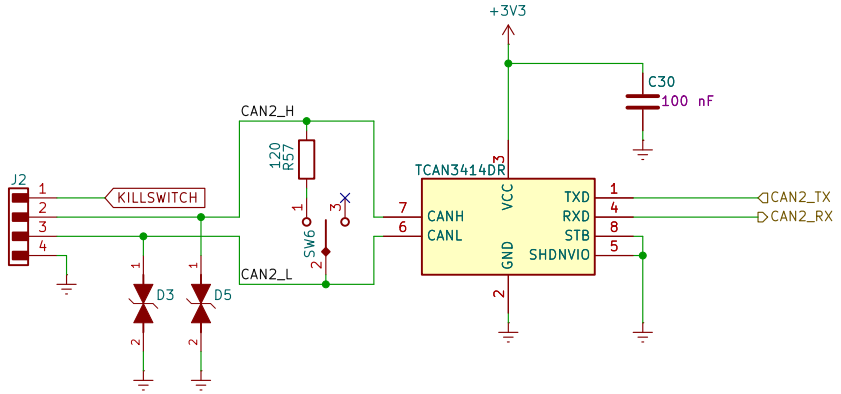
Size: A5	Date: 2026-10-14	**Rev:**
KiCad E.D.A. 10.0.1		Id: 1/4

V−Bat to +3V3



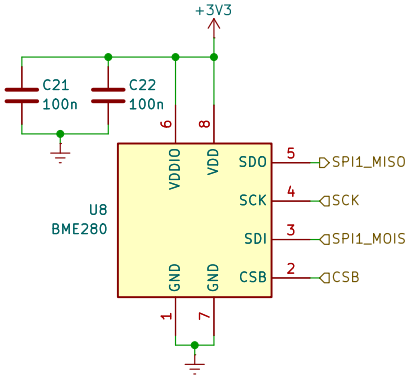
Values from datasheet, LED resistors calculated for 10mcd

TCAN



Values from datasheet
Switch used in testing to simulate tcan being last in chain

Telematic



Values from datasheet

Designed by: Simon Torvund

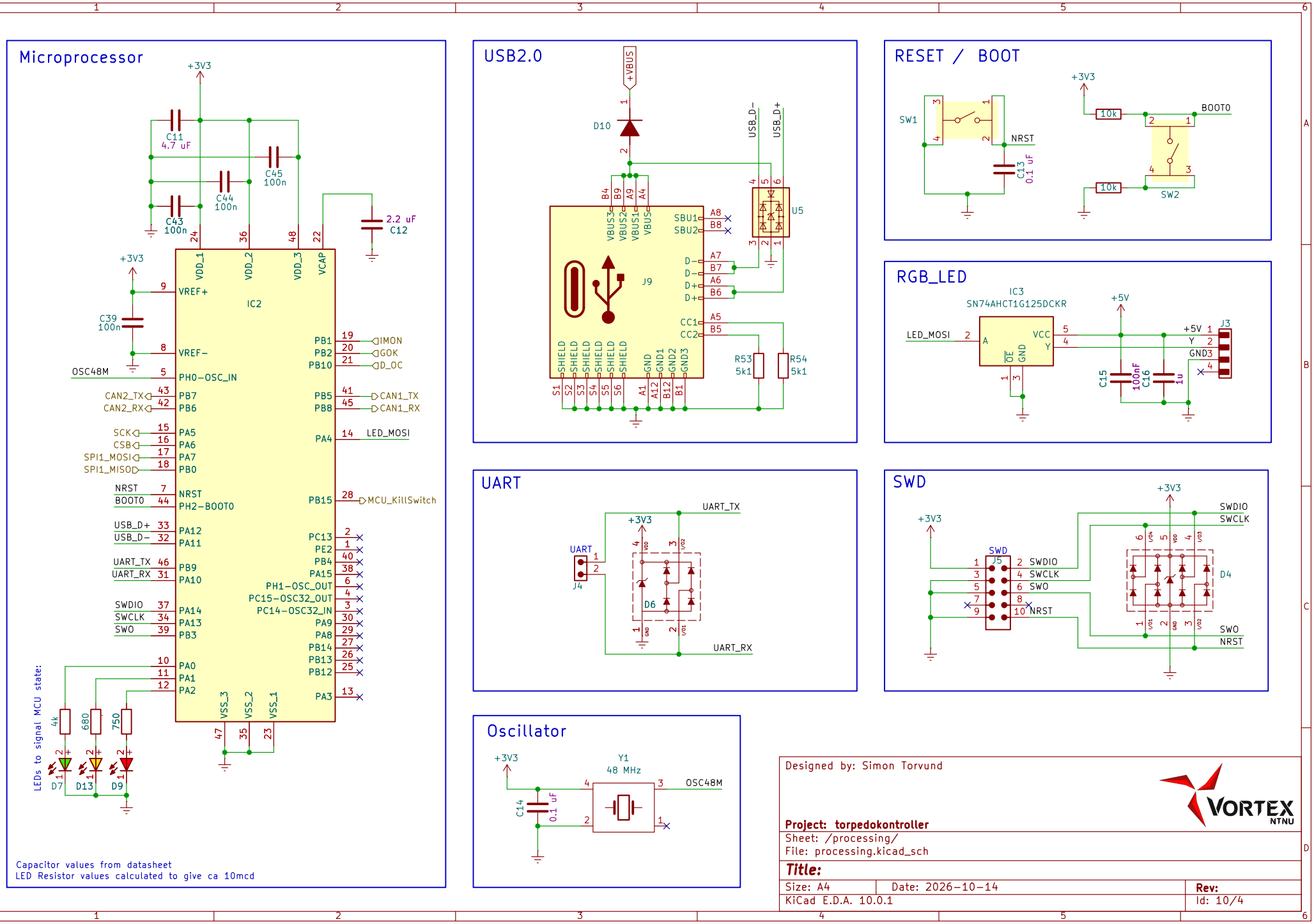
VORTEX NTNU

Project: torpedokontroller

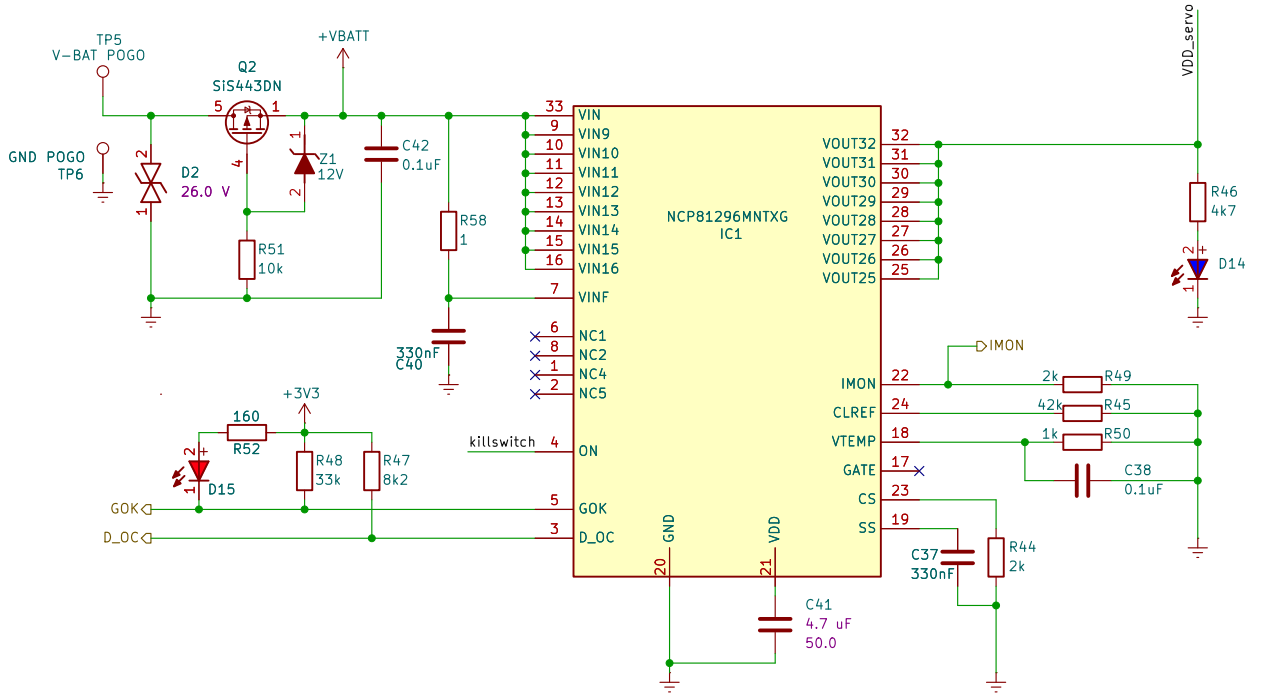
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File: input.kicad_sch

Title:

Size: A4	Date: 2026−10−14	**Rev:**
KiCad E.D.A. 10.0.1		Id: 9/4

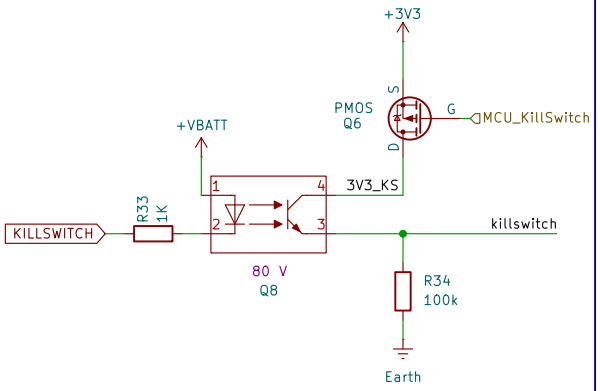

Microprocessor
+3V3
C11
4.7 uF
C45
100n
C44
100n
C43
100n
2.2 uF
C12
IC2
VDD_1
VDD_2
VDD_3
VCAP
VREF+
VREF−
PH0−OSC_IN
PB7
PB6
PA5
PA6
PA7
PB0
NRST
PH2−BOOT0
PA12
PA11
PB9
PA10
PA14
PA13
PB3
PA0
PA1
PA2
PB1
PB2
PB10
PB5
PB8
PA4
PB15
PC13
PE2
PB4
PA15
PH1−OSC_OUT
PC15−OSC32_OUT
PC14−OSC32_IN
PA9
PA8
PB14
PB13
PB12
PA3
VSS_3
VSS_2
VSS_1
C39
100n
OSC48M
CAN2_TX
CAN2_RX
SCK
CSB
SPI1_MOSI
SPI1_MISO
NRST
BOOT0
USB_D+
USB_D−
UART_TX
UART_RX
SWDIO
SWCLK
SWO
IMON
GOK
D_OC
CAN1_TX
CAN1_RX
LED_MOSI
MCU_KillSwitch
LEDs to signal MCU state:
4k
680
750
D7
D13
D9
Capacitor values from datasheet
LED Resistor values calculated to give ca 10mcd
USB2.0
+VBUS
D10
USB_D−
USB_D+
U5
J9
SBU1
SBU2
D−
D+
CC1
CC2
SHIELD
GND
GND1
GND2
GND3
R53
5k1
R54
5k1
RESET / BOOT
SW1
NRST
C13
0.1 uF
10k
BOOT0
SW2
RGB_LED
IC3
SN74AHCT1G125DCKR
+5V
LED_MOSI
VCC
OE
GND
J3
C15
100nF
C16
1u
UART
UART_TX
UART_RX
J4
D6
SWD
J5
SWDIO
SWCLK
SWO
NRST
D4
Oscillator
Y1
48 MHz
C14
0.1 uF
OSC48M
Designed by: Simon Torvund
VORTEX
NTNU
Project: torpedokontroller
Sheet: /processing/
File: processing.kicad_sch
Title:
Size: A4
Date: 2026−10−14
KiCad E.D.A. 10.0.1
Rev:
Id: 10/4

V-BAT TO VCC SERVO

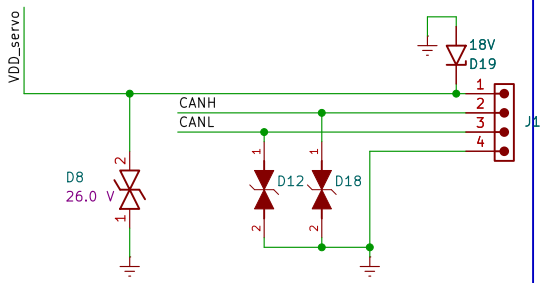


Values from datasheet and claude
Using SER-213X Servo by Blue trail, Operating Voltage 8.0v – 32v, no regulator needed

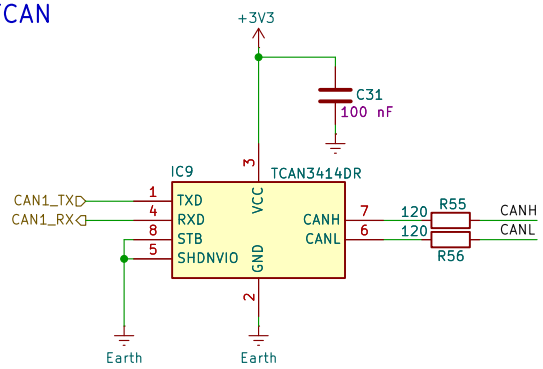
Killswitch



Servo connector



TCAN



Values from datasheet

Designed by: Simon Torvund

Project: torpedokontroller

Sheet: /output/
File: output.kicad_sch

Title:

Size: A4	Date: 2026-10-16	**Rev:**
KiCad E.D.A. 10.0.1		Id: 11/4